COMP34312 : Mathematical Topics in Machine Learning (Chapters 7 - 11)

Anirbit

Computer Science, UManchester

Plan

Chapter 7 (Set A) MCQs are VERY Important!

The concepts introduced in the Chapter 7 (Set A) quiz are REQUIRED to understand a LOT of the things that will be taught from here on.

Please let me know or come to my office hours on Thursdays, 1-2 PM @ Naval Suite, Crawford House to discuss anything that you might be unsure about that quiz — or anything in ML!

Notation for Function Classes

As you would have already realized...

In machine learning we need to think in terms of sets of functions/predictors/models i.e function classes.
So we need a consistent notation for this! We will illustrate our preferred notation via the set of all linear predictors in $d-$dimensions.

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$$\mathcal{F} = \left\{ \boldsymbol{x} \mapsto \langle \boldsymbol{w}, \boldsymbol{x} \rangle = \boldsymbol{w}^\top \boldsymbol{x} = \sum_{i=1}^{d} w_i x_i \mid w_i \in \mathbb{R} \;\; \forall i = 1, \ldots, d \right\}$$

(• read the symbol $\mapsto$ as “maps to”, $\mid$ as “for every”, $\forall$ as “for all” & $\in$ as “in”)

$\mathcal{F}$ above denotes the set of all linear functions acting on $d-$dimensionsal data ($\boldsymbol{x}$) with no restrictions on the weights $w_i, \; i = 1, \ldots, d$ i.e each w_i can be anything in $\mathbb{R}$.

An Interesting Example of a Function Class ($\mathcal{F}$) Nets Mapping $\mathbb{R}^4 \to \mathbb{R}^3$ Using The Following Architecture

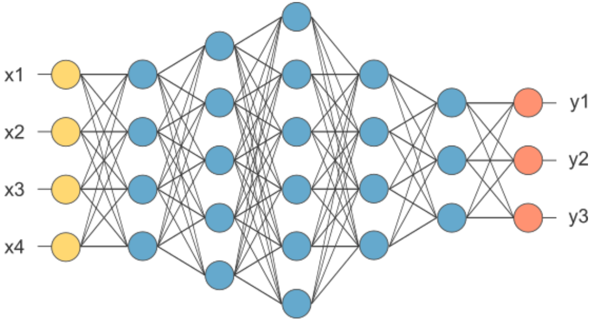

Figure 1: Note that in above the space of all "vectors" with $k-$coordinates is being denoted as $\mathbb{R}^k$. Such a neural diagram/architecture shown as a graph as above defines a certain set of neural functions corresponding to all possible weight assignments to the edges – and in the next slide we will see an explicit example.

An (Important) Example of a Loss Function Class ($\mathcal{H}$)

● Consider the following set of standard neural nets mapping $\mathbb{R}^2 \to \mathbb{R}$ using 2 gates/non-linearities,

$$\mathcal{F} = \left\{ (x,y) \mapsto \sum_{i=1}^{2} a_i \cdot \frac{1}{1 + e^{-(w_{i1}x + w_{i2}y)}} \mid a_1, a_2, w_{11}, w_{12}, w_{21}, w_{22} \in \mathbb{R} \right\}$$

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● Then the corresponding class of squared losses for a training data (x', y', z') is,

$$\mathcal{H} = \left\{ (a_1, a_2, w_{11}, w_{12}, w_{21}, w_{22}) \mapsto (z' - f(x', y'))^2 \mid f \in \mathcal{F} \right\}$$

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Notice a **key idea** here : that models/predictors are thought as functions of the input/data i.e (x, y), while losses are thought as functions of the weight/parameters of the models i.e $(a_1, a_2, w_{11}, w_{12}, w_{21}, w_{22})$.

Some Specific Properties of Functions Are Crucial To ML

Three key properties of differentiable functions that we shall care a lot about are, whether they are,

1. "Lipschitz" or not,
2. "Convex" or not
3. "Smooth" or not.

Which of these properties are true for a given loss function or model makes a dramatic difference to how machine learning algorithms behave for them!

Plan

[Concept 9] Introduction to the Euclidean Norm

Definition ($2-$Norm)

Given a vector $\boldsymbol{v} \in \mathbb{R}^p$ we define its $2-$Norm as,

$$\|\boldsymbol{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \ldots + v_p^2}$$

i.e the Euclidean distance of the point $\boldsymbol{v}$ from the origin.

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Eg. Consider the vector $v = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$**. Then it follows that**

$$\|\boldsymbol{v}\| = \sqrt{1^2 + (-2)^2 + (-1)^2} = \sqrt{6}$$

[Concept 8] Introduction to Lipschitzness

Definition (**Lipschitzness)**

A function $F : \mathbb{R}^p \to \mathbb{R}$ is said to be L–Lipschitz if $\forall\ \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^p$, we have,

$$|F(\boldsymbol{x}) - F(\boldsymbol{y})| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$$

[Concept 8] Introduction to Lipschitzness

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- But consider the function $F(x) = x^2$.

We can compare its values between say a point x and 0 and ask if it is true that there exists a $L > 0$ s.t for all x we will have, $|F(x) - F(0)| = x^2 \le L|x|$. But clearly for no $L > 0$ can the inequality $x^2 \le L|x|$ hold if $|x| > L$.

Thus $F(x) = x^2$ is not a Lipschitz function. **[Try Now?]**

[Concept 7] Introduction to Convexity

A convex function F can be informally understood as one whose graph between any two points x and y always lie below the straight line joining the points $(x, F(x))$ and $(y, F(y))$. *Note, that this notion of convexity does not need the function to be differentiable anywhere.*

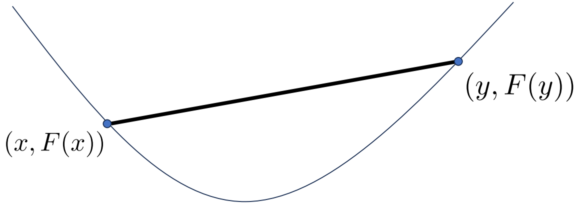

Figure 2: Convexity (Version 1)

[Concept 7] Introduction to Convexity

This view of convex function can be formally stated as,

Definition (**Convexity (Version 1)**)

A function $F : \mathbb{R}^p \to \mathbb{R}$ will be said to be a convex function if $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^p$ and $\forall \theta \in [0,1]$ we have,

$$F(\theta \boldsymbol{x} + (1-\theta)\boldsymbol{y}) \leq \theta F(\boldsymbol{x}) + (1-\theta)F(\boldsymbol{y})$$

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This above view of convexity will be extremely important in the last lecture of this course.

[Concept 7] Introduction to Convexity

But if the function can be assumed to be differentiable — as is the case for all optimization examples that we shall consider — then the following notion of convexity also becomes useful.

Definition (Convexity (Version 2))

An at least once differentiable function $F : \mathbb{R}^p \to \mathbb{R}$ is convex if, $\forall\ \boldsymbol{x}, \boldsymbol{y}$ we have,

$$F(\boldsymbol{x}) + \nabla F(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}) \leq F(\boldsymbol{y})$$

$$\text{where, } \nabla F(\boldsymbol{x}) \coloneqq \begin{bmatrix} \frac{\partial F}{\partial x_1} \\ \vdots \\ \frac{\partial F}{\partial x_p} \end{bmatrix} \Big|_{\boldsymbol{x}},$$

is the derivative of F that is assumed to be well-defined

[Concept 7] Introduction to Convexity

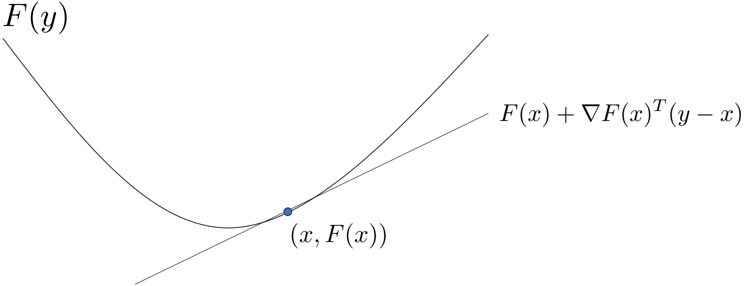
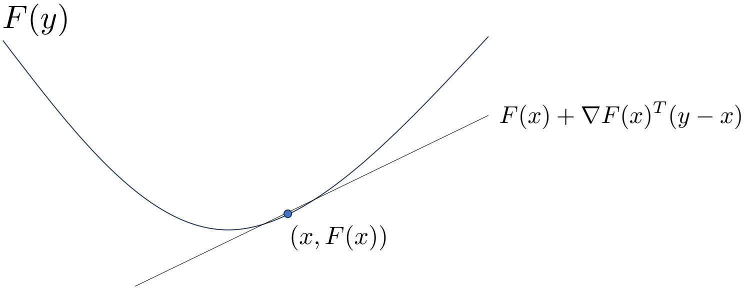


Figure 3: We can see that for differentiable functions, convexity can be thought of as the property that for any point x in the domain, the tangent to the function at that point is below the graph of the function.

Eg. $F(x) = x^2$, $F(x) = x^4$, $F(x) = e^{-x}$

[Concept 7] Introduction to Convexity



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Read the lecture notes for details on 2 special subsets of convex functions, — functions which satisfy, "strict convexity" & "strong convexity"

[Concept 7] Introduction to Convexity

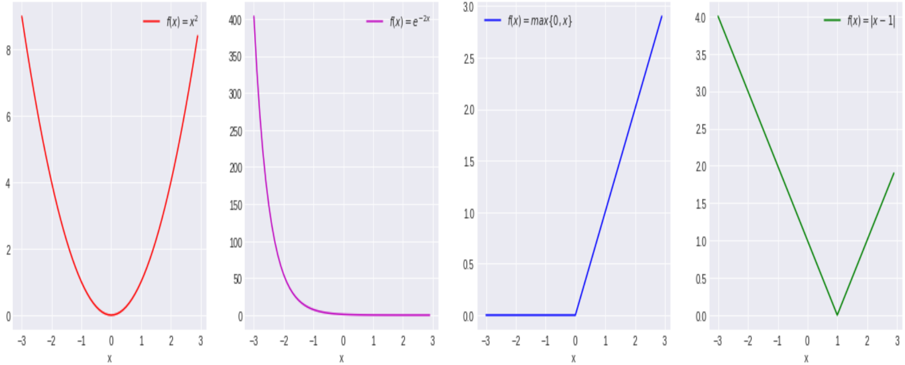

Figure 4: From left-to-right notice that we have examples of convex functions which (a) have a unique global minima (b) have no minima at all (c) are not differentiable everywhere and have an uncountable number of global minima and (d) are not differentiable everywhere but have a unique global minima.

[Concept 6] Introduction to Smoothness

Definition (**Lipschitz Smoothness)**

For some $\beta > 0$, an at least once differentiable function $F : \mathbb{R}^p \to \mathbb{R}$ is said to be β–smooth or β–Lipschitz smooth or β–gradient Lipschitz if its gradients are β–Lipschitz i.e $\forall\ \boldsymbol{x}, \boldsymbol{y}$, we have,

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Eg. $F(x) = x^2$ **is smooth with** $\beta = 2$**.** `Note that this terminology of ''smoothness'', differs from what is called a smooth function in calculus!`

In calculus, "smoothness" refers to a function being infinitely differentiable and we can see that **there are many examples of infinitely differentiable functions which are not smooth by the above definition, like $F(x) = x^3$ (Prove!)**.

Non-Convex Neural Losses

Way too many real-world loss functions have local minima which are not global. Now we shall see an example of such a situation - and in this example we shall use a strongly convex "regularizer" to emphasize that its presence alone cannot ameliorate this complexity of having non-trivial local minima.

Non-Convex Neural Losses

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Consider training data of 4 data, $(x_1, y_1) = (0.5, -100), (x_2, y_2) = (-1, 300), (x_3, y_3) = (1, 1), (x_4, y_4) = (-0.5, -400)$. Recall from earlier slides today, that one of the simplest neural nets is one consisting of a single sigmoid gate of weight w which would be mapping,

$$x \mapsto \frac{1}{1 + e^{-w \cdot x}} \in [0, \infty)$$

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$$w \mapsto F(w) \coloneqq \frac{1}{4}\sum_{i=1}^{4} \ell((x_i, y_i), w) + \underbrace{\lambda w^2}_{\text{"regularizer"}} \qquad (1)$$

$$\coloneqq \frac{1}{4}\sum_{i=1}^{4} \frac{1}{2} \cdot \left(y_i - \frac{1}{1 + e^{-w x_i}}\right)^2 + \lambda w^2 \in [0, \infty) \qquad (2)$$

Recall from earlier slides that we think of losses as functions of the weights.

Non-Convex Neural Losses

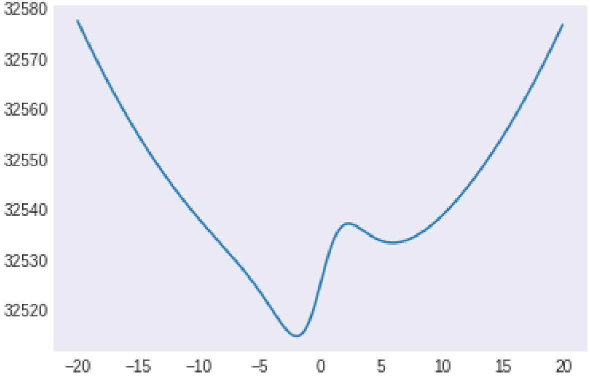

Figure 5: The plot of this F above, for $\lambda = 0.13$, as a function of w clearly shows that it is a non-Lipschitz function (owing to the regularizer) — and neither is it convex (since it has a local maxima) — and notice how it has two local minima, only one of which is a global minima slightly to the left of $w = 0$.

[Concept 5] Introduction to Convergence

To match usual conventions with optimization/training algorithms, we shall rename the weight variable as x. Thus we can re-write the above example as,

$$x \mapsto F(x) \coloneqq \frac{1}{4}\sum_{i=1}^{4} \ell((x_i, y_i), x) + 0.13 \cdot x^2 \tag{3}$$

$$\coloneqq \frac{1}{4}\sum_{i=1}^{4} \frac{1}{2} \cdot \left(y_i - \frac{1}{1 + e^{-x \cdot x_i}}\right)^2 + 0.13 \cdot x^2 \in [0, \infty) \tag{4}$$

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On the above loss function, an implementation of the **"Gradient Descent" (GD)** algorithm at a constant "step-length" of $\eta > 0$ and starting from x_0 would be an iterative execution of the following method,

$$x_{t+1} = x_t - \eta \cdot \frac{\mathrm{d}F}{\mathrm{d}x}\Big|_{x_t}$$

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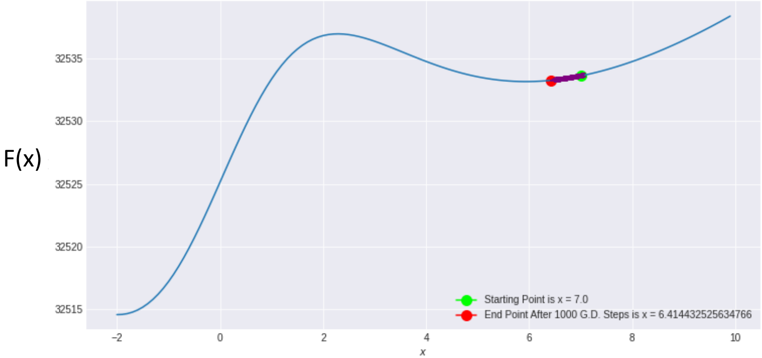

Figure 6: GD on F starting from $x_0 = 7$ and using a constant step-length of $\eta = 10^{-3}$ and running for 10^3 steps reaches $x \sim 6.414$

[Concept 5] Introduction to Convergence

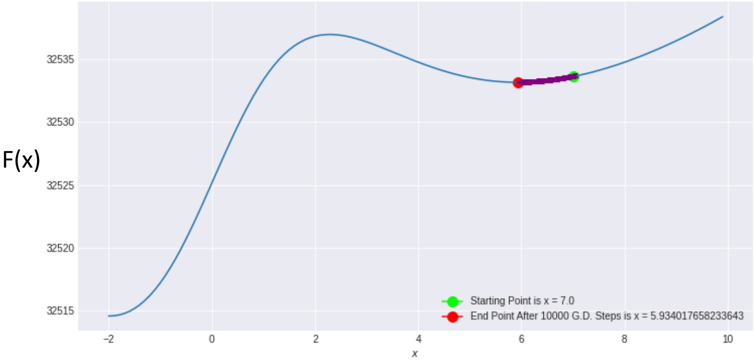

Figure 7: GD on F starting from $x_0 = 7$ and using a constant step-length of $\eta = 10^{-3}$ and running for 10^4 steps reaches $x \sim 5.934$

[Concept 5] Introduction to Convergence

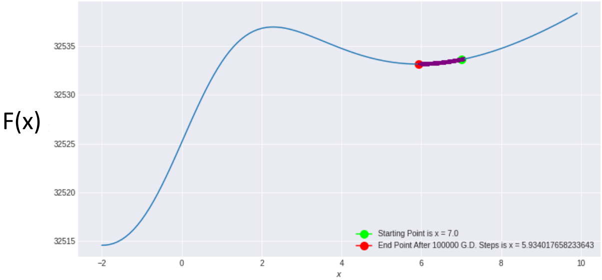

Figure 8: GD on F starting from $x_0 = 7$ and using a constant step-length of $\eta = 10^{-3}$ and running for 10^5 steps reaches $x \sim 5.934$

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It is clear from these figures that the algorithm makes significant progress between where it was after 10^3 steps and the next 9000 steps.

But for the next 90000 steps after $t = 10^4$, almost nothing seems to happen - the algorithm seems to have stalled. This is a tell-tale sign of "convergence" - that this sequence of numbers x_t seems to be getting arbitrarily close to a certain number as $t \to \infty$.

[Concept 5] Introduction to Convergence

Since here F was an actual example of a neural loss function – these experiments also point out that without a careful choice of starting point and step-lengths, GD can get "stuck" at a non-trivial local minima - and not reach the global minima of the function.

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From an earlier figure we know that this F indeed has a unique global minima, which the GD settings in the above diagrams seem to fail to reach.

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But can you find the right settings, when GD would reach the global minima?

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Definition (Convergence to a Limit Point)

A sequence of points $\boldsymbol{x}_i, \forall i = 1, 2, \ldots$ in $\mathbb{R}^n$, will be said to converge to a point $\boldsymbol{x}_* \in \mathbb{R}^n$ if $\lim_{n \to \infty} \|\boldsymbol{x}_n - \boldsymbol{x}_*\|_2 = 0$. If this condition is satisfied then $\boldsymbol{x}_*$ shall be called the "limit point" / "accumulation point" of the sequence.

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- The sequence $\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \ldots$ is a decreasing sequence, that converges to 0.
- The sequence $\left(-1 - \frac{1}{2}\right), \left(-1 - \frac{1}{3}\right), \left(-1 - \frac{1}{4}\right), \ldots$, is an increasing sequence that converges to the point -1

[Try Now?]

[Concept 3 & 4] Introduction to Supremums and Infimums

If one picks any $x \in [0,1)$ and wants to declare x as the maximum of this interval then one can always come up a very small number $\epsilon > 0$ s.t $x + \epsilon \in [0,1)$ and thus $x + \epsilon$ becomes a more credible candidate for being called the maximum - and starting from $x + \epsilon$ we can again repeat this argument and this can be continued ad infinitum!

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Let's Think More Carefully!

[Concept 3 & 4] Introduction to Supremums and Infimums

For the case $I = [0, 1)$ considered earlier realize that it has uncountably infinite number of upper and lower bounds.

(a) Every real number greater that or equal to 1 is an "upper bound" on I,
- because all numbers in I are less than or equal to any number ≥ 1
- and there are an uncountably infinite number of them.

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Can we make these bounding numbers unique in some way while side-stepping the problem with maximums and minimums ?

[Concept 3 & 4] Introduction to Supremums and Infimums

Definition (Supremum)

Let I be a subset of $\mathbb{R}$. Suppose there exists an upper bound M on I s.t $M \leq M'$, $\forall$ upper bounds M' on I. Then M is the "least upper bound" on I and it's called the "supremum" of I and is denoted as $\sup I$.
(And when $\sup I \in I$ we call it the "maximum" of I, $\max I$.)

[Concept 3 & 4] Introduction to Supremums and Infimums

Definition (Supremum)

Let I be a subset of $\mathbb{R}$. Suppose there exists an upper bound M on I s.t $M \leq M'$, $\forall$ upper bounds M' on I. Then M is the "least upper bound" on I and it's called the "supremum" of I and is denoted as $\sup I$. *(And when $\sup I \in I$ we call it the "maximum" of I, $\max I$.)*

Definition (Infimum)

Let I be a subset of $\mathbb{R}$. Suppose there exists a lower bound m on I s.t $m' \leq m$, $\forall$ lower bounds m' on I. Then m is the "greatest lower bound" on I and it's called the "infimum" of I and is denoted as $\inf I$. *(And when $\inf I \in I$ we call it the "minimum" of I, $\min I$.)*

For intuition, look at the previous example again to realize that (a) when $I = [0, 1)$, we have $\sup I = 1$ and $\inf I = 0 = \min I$.

[Concept 2] Introduction to Spectral Norms of Matrices

Definition

For any matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ we define its $2,2$-norm as,

$$\|\boldsymbol{A}\|_{2,2} \coloneqq \sup_{\|\boldsymbol{x}\|_2=1} \|\boldsymbol{A}\boldsymbol{x}\|_2 = \sup_{\boldsymbol{x}\neq 0} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2}$$

When its clear from context, the above will also be called just the $2-$norm of $\boldsymbol{A}$ and denoted as $\|\boldsymbol{A}\|_2$ or spectral norm of $\boldsymbol{A}$.

[Concept 2] Introduction to Spectral Norms of Matrices

Let's See A Tricky Example Of It! :D

[Concept 2] Introduction to Spectral Norms of Matrices

Let's See A Tricky Example Of It! :D

Consider the following family of matrices,

$$\boldsymbol{A}(\theta) \coloneqq \begin{bmatrix} 0 & \theta \\ 0 & 0 \end{bmatrix}$$

One can solve the equation $\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}$ to obtain the eigenvalues (λ) and eigenvectors $(\boldsymbol{v})$ for this matrix and realize that its only eigenvalue is 0.

[Concept 2] Introduction to Spectral Norms of Matrices

But, for an arbitrary vector on the unit circle in $\mathbb{R}^2$

- parameterized as say $\begin{bmatrix} \sin(\alpha) \\ \cos(\alpha) \end{bmatrix}$ we have,

$$\left\| \boldsymbol{A}(\theta) \begin{bmatrix} \sin(\alpha) \\ \cos(\alpha) \end{bmatrix} \right\| = \left\| \begin{bmatrix} \theta \cos(\alpha) \\ 0 \end{bmatrix} \right\| = |\theta| \cdot |\cos(\alpha)|$$

Thus we have,

$$\| \boldsymbol{A}(\theta) \| = \sup_{\alpha} |\theta| \cdot |\cos(\alpha)| = |\theta|$$

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So, by choosing θ arbitrarily large in magnitude we can make spectral norm of $\boldsymbol{A}(\theta)$ as large as we want, *while the largest eigenvalue magnitude (also called the "spectral radius" of a matrix) remains at* 0, $\forall \theta$. **[Try Now?]**

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In general, one can show,

spectral radius $\leq$ spectral norm

– but the gap between them can be arbitrarily large!

[Concept 1] Introduction to Rank of Matrices

Now, consider the following 3 matrices,

$$\boldsymbol{A} \coloneqq \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \boldsymbol{B} \coloneqq \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \boldsymbol{C} \coloneqq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

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For any arbitrary vector $\boldsymbol{v} \coloneqq \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2$,

we have its image under the matrices being,

$$\boldsymbol{A}[\boldsymbol{v}] = 2v_1\boldsymbol{e}_1 + v_2\boldsymbol{e}_2,\ \boldsymbol{B}[\boldsymbol{v}] = -v_1\boldsymbol{e}_1,\ \boldsymbol{C}[\boldsymbol{v}] = 0$$

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Relating back to Chapter 7 (Set A) quiz we can observe,

$$\text{dimension}\left(\{\boldsymbol{A}[\boldsymbol{v}] \mid \boldsymbol{v} \in \mathbb{R}^2\}\right) = 2,\ \text{dimension}\left(\{\boldsymbol{B}[\boldsymbol{v}] \mid \boldsymbol{v} \in \mathbb{R}^2\}\right) = 1$$

$$\text{dimension}\left(\{\boldsymbol{C}[\boldsymbol{v}] \mid \boldsymbol{v} \in \mathbb{R}^2\}\right) = 0$$

[Concept 1] Introduction to Rank of Matrices

The calculation above can be said to prove that that the **rank of the matrices A, B & C are $2, 1$ & 0 respectively**. *It is easy to see how to extend this discussion to arbitrary rectangular matrices.*

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Definition (**Image and Rank of a Matrix)**

Given any $m \times n$ matrix $\boldsymbol{T}$ i.e $\boldsymbol{T} : \mathbb{R}^n \to \mathbb{R}^m$, we define its "image" as,

$$\mathrm{Image}(\boldsymbol{T}) \coloneqq \{\boldsymbol{T}[\boldsymbol{v}] \mid \boldsymbol{v} \in \mathbb{R}^n\}$$

. Then we can define,

$$\mathrm{rank}(\boldsymbol{T}) \coloneqq \mathrm{dimension}(\mathrm{Image}(\boldsymbol{T}))$$

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The calculation above can be said to prove that that the **rank of the matrices A, B & C are $2, 1$ & 0 respectively**. *It is easy to see how to extend this discussion to arbitrary rectangular matrices.*

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. Then we can define,

$$\text{rank}(\boldsymbol{T}) \coloneqq \text{dimension}(\text{Image}(\boldsymbol{T}))$$

- **If $\text{rank}(\boldsymbol{T}) = \min\{n, m\}$, then we call T to be of "full rank".**

Plan

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In the real world, for training the large neural net models, the algorithms that are used are far more complex – than just taking gradient steps.

But to really appreciate the training technologies in use at the bleeding edge of modern data-science, we need to start from the most vanilla form of this algorithm i.e **"Gradient Descent" (GD)**

[Non-Assessed] See, Sections 12.7 to 12.11 of "d2l.ai" for the various kinds of gradient based training methods that are most successful on modern massively over-parameterized neural nets.

What is GD?

GD on a Differentiable Function $F : \mathbb{R}^p \to \mathbb{R}$

1: **Input:** An initial point $\boldsymbol{w}_0$ and a specification $T > 0$ of the number of steps for the GD
2: **Choose :** A step-size sequence $\eta_t > 0, \ \forall t = 0, 1, 2, \ldots.$
3: **for** $t = 0, \ldots, T - 1$ **do**
4: $\quad \boldsymbol{w}_{t+1} = \boldsymbol{w}_t - \eta_t \cdot \nabla F(\boldsymbol{w}_t)$
5: **end for**
6: **Output :** $\boldsymbol{w}_T$

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The Key Question - Largely Unresolved!

"If F has global minima then when does GD find one of them,
– and which one and how fast?"

Introduction to Proof of Convergence of GD

Recall the idea of convergence of GD from the last class. In the following theorem we shall see an example of making that precise for a simple objective function $F(x) = x^2$
- that is strongly convex, differentiable and Lipschitz smooth.

Introduction to Proof of Convergence of GD

Recall the idea of convergence of GD from the last class. In the following theorem we shall see an example of making that precise for a simple objective function $F(x) = x^2$
- that is strongly convex, differentiable and Lipschitz smooth.

For this objective function, we can see that the GD steps are,

$$\begin{aligned} x_{t+1} &= x_t - \underbrace{\eta_t}_{\text{step–length}} F'(x_t) \\ &= x_t - \eta_t \cdot 2x_t \\ &= (1 - 2\eta_t)x_t \end{aligned}$$

Introduction to Proof of Convergence of GD

$$x_{t+1} = x_t - \underbrace{\eta_t}_{\text{step−length}} F'(x_t) = x_t - \eta_t \cdot 2x_t = (1 - 2\eta_t)x_t$$

[Try Now?]

Theorem (**Provable Rate of GD Convergence on x^2**)

Consider doing GD on the function $F(x) = x^2$, starting from any $x_0 \in \mathbb{R}$, $x_0 \neq 0$ and using any choice of constant step-length of $\frac{1-k}{2}$ for any $k \in (0,1)$. Then, (a) the iterates will asymptotically in time converge to the global minima at $x = 0$

&

(b) taking $\frac{\log\left(\frac{|x_0|}{\varepsilon}\right)}{\log \frac{1}{k}}$ number of steps is sufficient for the iterates to be within a distance of ε of the global minima, for all $\varepsilon > 0$.

Proof of Convergence of GD on x^2

Proof.

- Lets "unroll" $x_{t+1} = (1 - 2\eta_t)x_t$, to get,

$$
\begin{aligned}
x_{t+1} &= (1 - 2\eta_t)x_t = (1 - 2\eta_t)(1 - 2\eta_{t-1})x_{t-1} \\
&= (1 - 2\eta_t)(1 - 2\eta_{t-1})\ldots(1 - 2\eta_{t-t})x_{t-t} = x_0 \prod_{i=0}^{t}(1 - 2\eta_i)
\end{aligned}
$$

$$\implies x_t = x_0 \prod_{i=0}^{t-1}(1 - 2\eta_i)$$

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- Lets observe a few things about this above equation!
  - If we had $x_0 = 0$ then it implies that for all times t, we would have $x_t = 0$. Thus the algorithm would not move at all. This is why it was important to assume in the theorem that $x_0 \neq 0$.

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- Lets observe a few things about this above equation!
  - If we had $x_0 = 0$ then it implies that for all times t, we would have $x_t = 0$. Thus the algorithm would not move at all. This is why it was important to assume in the theorem that $x_0 \neq 0$. More generally if the algorithm were to start at a point x_0 s.t $F'(x_0) = 0$ i.e x_0 is a "critical point" of F, then the algorithm would not move at all. **Thus critical points are not to be used as starting points of GD algorithms.**

Proof of Convergence of GD on x^2

Proof.

- We had "unrolled" $x_{t+1} = (1 - 2\eta_t)x_t$, to get, $x_t = x_0 \prod_{i=0}^{t-1}(1 - 2\eta_i)$
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- Lets observe a few things about this above equation!
  - We want $x_t \to 0$ (the global minima of F) as $t \to \infty$,
    & from above it follows that a simple sufficient condition for that to happen is if for some constant $k \in (0,1)$, $(1 - 2\eta_i) = k, \ \forall i$
    — and this can be ensured by choosing a constant step-length as,

$$\eta_t = \frac{1}{2} \cdot (1 - k)$$

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- Making the above choice we have, $(1 - 2\eta_i) = k, \ \forall i$, and we can rewrite what we previously had for x_t as, $x_t = x_0 \prod_{i=0}^{t-1}(k) = x_0 \cdot k^t$

Proof of Convergence of GD on x^2

Proof.

- Thus we have,

$$\lim_{t\to\infty} x_t = \lim_{t\to\infty} x_0 \cdot k^t = 0$$

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- To obtain convergence time ("non-asymptotic") bounds, we check how long it takes to get within a $\varepsilon > 0$ interval of the global minima.

$$|x_t| \leq \varepsilon \implies |x_0| \cdot k^t \leq \varepsilon \implies t \geq \frac{\log\left(\frac{|x_0|}{\varepsilon}\right)}{\log \frac{1}{k}}$$

[Try Now?]

Proof of Convergence of GD on x^2

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Thus we have arrived at the fact that we set out to prove.

This is Only the Beginning!

One of the key insights in the theory of GD is that one of the basic ways in which a function gets non-trivial for GD to work with is if the objective is **either non-convex or non-Lipschitz smooth**
– *as is almost always the case in the real world!*.

This is Only the Beginning!

One of the key insights in the theory of GD is that one of the basic ways in which a function gets non-trivial for GD to work with is if the objective is **either non-convex or non-Lipschitz smooth**
– *as is almost always the case in the real world!*.

Thus we need to start to go beyond the confines of smooth convex functions.

Demonstration of GD convergence, on a differentiable convex function which is – neither Lipschitz nor Lipschitz smooth.

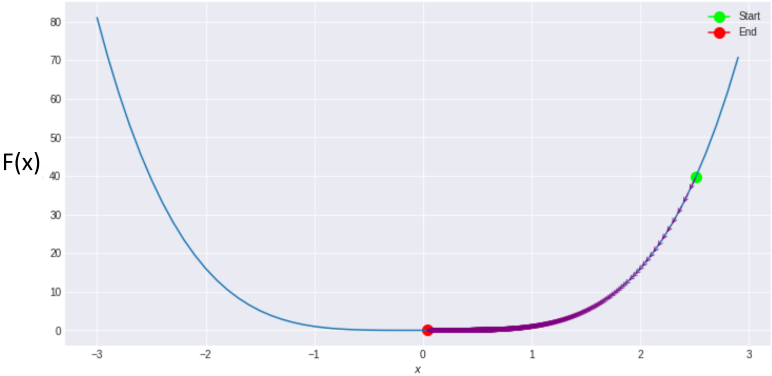

Figure 9: Progress to the unique global minima of GD on the function $F(x) = x^4$ for step-size $\eta = 10^{-3}$ and $T = 10^5$ and the algorithm starting near 2.5

Demonstration of GD convergence, on a differentiable function which is – neither convex, nor Lipschitz nor Lipschitz smooth.

This instance is more complex than the previous one - the target function here is not even convex! (Coming Up in Practice!)

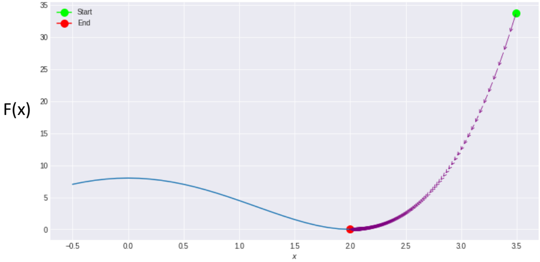

Figure 10: Progress to a global minima of G.D. on the function $F(x) = \frac{1}{2} \cdot (x^2 - 2^2)^2$ for $\eta = 10^{-3}$ and $T = 10^4$ and the algorithm starting near 3.5

Glimpses of the Road Beyond this Course

The situations as in this Mexican-Hat example above *are not at all rare* – in ways this is reflective of some of the key features of how actual deep-learning happens.

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Thus we saw two quite varied examples where this deceptively simple looking gradient descent algorithm seems to be finding the global minima and also while using constant step-sizes.

How does one prove convergence in such increasingly hard cases?

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Thus we saw two quite varied examples where this deceptively simple looking gradient descent algorithm seems to be finding the global minima and also while using constant step-sizes.

How does one prove convergence in such increasingly hard cases?

(Do a PhD!)

Plan

What is Linear Regression?

What is Linear Regression?

Consider being given n data points in d dimensions, stacked as a $n \times d$ matrix called $\boldsymbol{X}$ — whose (i,j)–entry would be denoted as X_{ij}. Let the corresponding "labels" be given as a column vector $\boldsymbol{y} \in \mathbb{R}^{n\times 1}$ — whose i^{th}–entry would be denoted as y_i.

We focus on the "linear regression" optimization question which can be stated as,

$$\min_{\boldsymbol{\beta}\in\mathbb{R}^d} \quad \underbrace{\frac{1}{2}\left\| \underbrace{\boldsymbol{y}}_{n\times 1} - \underbrace{\boldsymbol{X}}_{n\times d}\underbrace{\boldsymbol{\beta}}_{d\times 1} \right\|_2^2}_{\text{empirical risk of linear regression}}$$

What is Linear Regression?

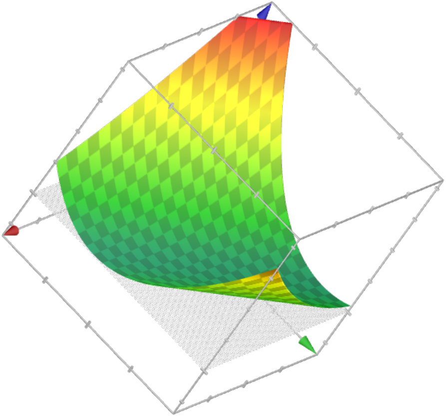

Figure 11: A plot of the empirical risk of an instance of "over-parameterized" linear regression in $2-$variables with one data, $y = 4$, $\boldsymbol{X} = [1, 2]$. The red marked axis on the left is for β_1 and the green marked axis on the right is for β_2

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This motivates us to define the empirical risk of linear regression, $\hat{R}$ as,

$$\hat{R}(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 \tag{5}$$

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Constant Step-Size GD attempting to solve this would schematically be,

1: **Input:** A starting point for the algorithm, $\boldsymbol{\beta}_0 \in \mathbb{R}^d$
2: **Input:** A step-size $\eta > 0$.
3: **for** $t = 1, \ldots$ **do**
4: $\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} - \eta \nabla \hat{R}(\boldsymbol{\beta}_{t-1})$ $\qquad \triangleright \ \frac{\partial \hat{R}}{\partial \boldsymbol{\beta}} := \begin{bmatrix} \frac{\partial \hat{R}}{\partial \beta_1} \\ \vdots \\ \frac{\partial \hat{R}}{\partial \beta_d} \end{bmatrix} = -\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$
5: **end for**

What Does Constant Step-Size GD do - on Linear Regression?

Keep Repeating This Update Rule - But Where is it Headed?

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} - \eta \nabla \hat{R}(\boldsymbol{\beta}_{t-1}) \qquad \triangleright \; \frac{\partial \hat{R}}{\partial \boldsymbol{\beta}} := \begin{bmatrix} \frac{\partial \hat{R}}{\partial \beta_1} \\ \vdots \\ \frac{\partial \hat{R}}{\partial \beta_d} \end{bmatrix} = -\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$$

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We assume two things about this setup - for realism and tractability,

What Does Constant Step-Size GD do - on Linear Regression?

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We assume two things about this setup - for realism and tractability,

- We assume that the model is "overparameterized", i.e n (#training data) $< d$ (#training parameters).
- The data matrix $\boldsymbol{X}$ is full rank. (i.e $\boldsymbol{X}$ has rank n while $n < d$)

A Full-Rank Data Assumption Helps Analysis

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If $\boldsymbol{X} \in \mathbb{R}^{n \times d}$ *is of rank* n *and* $n < d$ *then* $\boldsymbol{X}\boldsymbol{X}^\top \in \mathbb{R}^{n \times n}$ *is invertible.*

- We will skip the proof of this.

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- We will skip the proof of this.
- But understand that inverse of a square matrix $\boldsymbol{A}$ is denoted as $\boldsymbol{A}^{-1}$ and its s.t $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$.
  ($\boldsymbol{I}$ has ones on its diagonal and zero at all other entries)
- Invertibility is not guaranteed and it only happens, if and only if, a matrix does not transform non-zero vectors to zero vectors. Eg. matrices $\boldsymbol{B}$ and $\boldsymbol{C}$ from Concept-1 were not invertible but $\boldsymbol{A}$ was. **[Try Now?]**

Where are the Global Minima of Linear Regression?

Given the above realization, we can conclude that at $\boldsymbol{\beta} = \boldsymbol{X}^\top(\boldsymbol{X}\boldsymbol{X}^\top)^{-1}\boldsymbol{y}$ our loss function $\hat{R} = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2$ is at its global minima i.e 0.
So, we denote,

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But is this $\boldsymbol{\beta} = \boldsymbol{X}^\dagger\boldsymbol{y}$ the only global minima of $\hat{R}$? No!

In the next claim we shall realize how there are multiple global minima of $\hat{R}$.

Lemma

Let $\boldsymbol{x}_i^\top \in \mathbb{R}^{1\times d}$ be the $n(< d)$ rows of the data matrix $\boldsymbol{X}$. Given $\boldsymbol{y}$ and $\boldsymbol{X}$ as above, $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2 = 0$ iff $\boldsymbol{\beta} = \boldsymbol{X}^\dagger\boldsymbol{y} + \boldsymbol{\xi}$ where $\boldsymbol{\xi}$ is s.t $\boldsymbol{\xi}^\top\boldsymbol{x}_i = 0$ for all i.

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Proof. Given $\boldsymbol{\beta}$ as above we note that,

$$\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\| = \|\boldsymbol{y} - \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\xi}\| = \|\boldsymbol{X}\boldsymbol{\xi}\| = 0$$

The last equality follows since it has been assumed that every row of $\boldsymbol{X}$ is orthogonal to $\boldsymbol{\xi}$. **Conversely,** suppose $\boldsymbol{\beta}$ is s.t $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\| = 0$. Then we can define $\boldsymbol{\eta} \coloneqq \boldsymbol{\beta} - \boldsymbol{X}^\dagger \boldsymbol{y}$ and it follows that $\|\boldsymbol{X}\boldsymbol{\eta}\| = 0$.

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Where are the Global Minima of Linear Regression?

But the solution $\boldsymbol{X}^{\dagger}\boldsymbol{y}$ to linear regression is still special and that is because it is the global minima of our loss with the least norm — i.e the solution of the linear regression problem that lies closest to the origin in the parameter space.

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Lemma

$$\boldsymbol{X}^{\dagger}\boldsymbol{y} = \underset{\boldsymbol{\beta} \text{ s.t } \hat{R}(\boldsymbol{\beta})=0}{\operatorname{argmin}} \|\boldsymbol{\beta}\|_2$$

Where are the Global Minima of Linear Regression?

Proof.

From the previous claim we know that if any $\boldsymbol{\beta}$ is s.t $\hat{R}(\boldsymbol{\beta}) = 0$ then we can write it as $\boldsymbol{\beta} = \boldsymbol{X}^{\dagger}\boldsymbol{y} + \boldsymbol{\xi}$ s.t $\boldsymbol{\xi}$ is orthogonal to each of the rows of $\boldsymbol{X}$.

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$$\begin{aligned}
\|\boldsymbol{\beta}\|^2 &= \left\|\boldsymbol{X}^{\dagger}\boldsymbol{y}\right\|^2 + \|\boldsymbol{\xi}\|^2 + 2\left\langle \boldsymbol{X}^{\dagger}\boldsymbol{y}, \boldsymbol{\xi}\right\rangle \\
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$$\|\boldsymbol{\beta}\|^2 = \left\|\boldsymbol{X}^{\dagger}\boldsymbol{y}\right\|^2 + \|\boldsymbol{\xi}\|^2 \geq \left\|\boldsymbol{X}^{\dagger}\boldsymbol{y}\right\|_2^2 \qquad [\textbf{TryNow}?] \tag{8}$$

In the last step above its clear that the equality holds only when $\boldsymbol{\xi} = 0$ and thus we have proven that the $2-$norm of all global minimizers of our linear regression loss $\hat{R}$ are bigger than that of $\boldsymbol{X}^{\dagger}\boldsymbol{y}$.

□

Which Global Minima of $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2$ Does GD Find?

Now lets consider doing G.D. on $\hat{R}(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2$ starting from an initial weight $\boldsymbol{\beta}_0$ and using a constant step-length $\eta > 0$,

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} - \eta \nabla \hat{R}(\boldsymbol{\beta}_{t-1})$$

We note the following preliminary observation,

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If the data matrix $\boldsymbol{X}$ is full rank, $\exists$ a continuum of choices of η s.t GD converges to a global minimizer of $\hat{R}$. **(Proof of this is NOT in the Syllabus - References are Given in the Lecture Notes to Help Reconstruct the Full Proof)**

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But we saw just now that with full rank data, there are uncountably many global minima for linear regression!

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Theorem

(Main Result) Full-Rank Overparameterized Linear Regression by GD Can Be Setup s.t. It Finds The Minimum $2-$Norm Interpolant *If the data matrix $\boldsymbol{X}$ is full rank and $\boldsymbol{\beta}_0 = 0$ then $\exists$ a continuum of choices of step-lengths η at which GD on the overparameterized linear regression loss $\hat{R}(\boldsymbol{\beta}) = \frac{1}{2} \cdot \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2$ converges to the global minimizer with the minimum $2-$norm i.e $\boldsymbol{X}^\dagger \boldsymbol{y}$.*

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Proof.

Recall that we have denoted $\mathbf{x}_1^\top, \ldots, \mathbf{x}_n^\top$ as the rows of the data matrix $\boldsymbol{X}$ and each $\mathbf{x}_i \in \mathbb{R}^{d\times 1}$.

[KEY IDEA] Via induction, we shall first try to prove that any of the iterates of this GD is a linear combination of the data. □

Which Global Minima of $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2$ Does GD Find?

Proof.

To start the induction, note that trivially we have, $\boldsymbol{\beta}_0 = 0 \in \mathrm{Span}(\mathbf{x}_1, \ldots, \mathbf{x}_n)$. Next, suppose that, $\boldsymbol{\beta}_{t-1} \in \mathrm{Span}(\mathbf{x}_1, \ldots, \mathbf{x}_n)$. Now recall that we had,

$$\hat{R}(\boldsymbol{\beta}) = \frac{1}{2} \cdot \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 = \frac{1}{2} \cdot \sum_{i=1}^{n} (y_i - (\boldsymbol{X}\boldsymbol{\beta})_i)^2 = \frac{1}{2} \cdot \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{d} X_{ij} \boldsymbol{\beta}_j \right)^2$$

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So, we get, $\frac{\partial \hat{R}}{\partial \beta_j} = -\sum_{i=1}^{n} (y_i - \sum_{k=1}^{d} X_{ik} \boldsymbol{\beta}_k) \cdot X_{ij}$. Stacking these into a column vector of d dimensions we have the derivative of the loss function w.r.t the weight vector being writable succinctly as,

$$\frac{\partial \hat{R}}{\partial \boldsymbol{\beta}} := \begin{bmatrix} \frac{\partial \hat{R}}{\partial \beta_1} \\ \vdots \\ \frac{\partial \hat{R}}{\partial \beta_d} \end{bmatrix} = -\boldsymbol{X}^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$$

□

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Proof.

Recall that the i^{th}–column of $\boldsymbol{X}^\top$ is the vector $\mathbf{x}_i$. So the matrix multiplication that is done in the above expression of $\frac{\partial \hat{R}}{\partial \boldsymbol{\beta}}$ can be re-written as the following weighted sum of vectors,

$$\frac{\partial \hat{R}}{\partial \boldsymbol{\beta}} = \sum_{i=1}^{n} (-y_i + (\boldsymbol{X}\boldsymbol{\beta})_i) \cdot \mathbf{x}_i$$

Hence specific to $\boldsymbol{\beta}_{t-1}$, we can conclude from above that, $\frac{\partial \hat{R}}{\partial \boldsymbol{\beta}_{t-1}} \in \mathrm{Span}(\mathbf{x}_1, \ldots, \mathbf{x}_n)$. Further, from the update rule being, $\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} - \eta \nabla \hat{R}(\boldsymbol{\beta}_{t-1})$ and recalling the induction hypothesis we can conclude what we set out to prove, that,

$$\boldsymbol{\beta}_t \in \mathrm{Span}(\mathbf{x}_1, \ldots, \mathbf{x}_n), \ \forall t = 1, 2, \ldots$$

$\square$

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Note that, $\hat{\boldsymbol{\beta}} \in \mathrm{Span}(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is equivalent to saying that $\exists\ \boldsymbol{v} \in \mathbb{R}^n$ s.t $\hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \boldsymbol{v}$.

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$$\hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \underbrace{(\boldsymbol{X}\boldsymbol{X}^\top)^{-1}\boldsymbol{y}}_{\boldsymbol{v}} = \boldsymbol{X}^\dagger \boldsymbol{y} \implies \hat{R}(\hat{\boldsymbol{\beta}}) = 0.$$

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Thus by considering the the same weight that was previously shown to be the minimum norm global minimizer of $\hat{R}$ (an "interpolant" since $\hat{R} = 0$ at this weight), we realize that $\exists\ \hat{\boldsymbol{\beta}} \in \mathrm{Span}(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ s.t $\hat{R}(\hat{\boldsymbol{\beta}}) = 0$.

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Towards showing its uniqueness – suppose we have a $\hat{\boldsymbol{\beta}} \in \mathrm{Span}(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ (i.e $\exists\ \boldsymbol{v} \in \mathbb{R}^n$ s.t $\hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \boldsymbol{v}$) s.t $\hat{R}(\hat{\boldsymbol{\beta}}) = 0$. Then,

$\left\|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}\right\| = 0 \implies \boldsymbol{y} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{v}$.

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Which Global Minima of $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2$ Does GD Find?

Proof.

Thus we have shown that there exists a unique $\hat{\boldsymbol{\beta}}$ in the span of the data such that $\hat{R}(\hat{\boldsymbol{\beta}}) = 0$. Since 0 is the lowerbound on $\mathcal{L}$, *it follows that our linear regression loss has a unique global minima in the span of its data*.

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□

Can you instantiate GD experiments on linear regression that converge to a global minima but which is not this min-norm solution?

Looking Ahead!

Thus we have seen an explicit example of how having a large model size compared to the number of data i.e "overparameterization" leads to a multiplicity of global minima for the loss function — and a surprising selectivity to creep into the GD algorithm as to which global minima it will converge to.

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Hence Research! :D

Semester 2 Surveys!

Course Unit Surveys 2025/2026 - Semester 2

Plan

Learning Capacity of a Function Class

Recall that we have learnt that the "population risk" function measures the "out of sample" performance of a predictor - and its definition depends on three key components, **(a)** the data distribution, **(b)** the loss function and **(c)** the predictor being used.

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Recall that we had defined $f : \mathbb{R}^d \to \mathbb{R}^k$ as the predictor (the model at hand, say a neural net) and the loss function $\ell : \mathbb{R}^k \times \mathbb{R}^k \to [0, \infty)$ and the data $(\boldsymbol{x}, \boldsymbol{y}) \in \mathbb{R}^d \times \mathbb{R}^k$ sampled from a data distribution say $\mathcal{D}$ on $\mathbb{R}^d \times \mathbb{R}^k$. Towards emphasizing certain specific dependencies in the risk function we shall from now denote the "population risk" of f as,

$$R(f) \coloneqq \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})} \left[\ell(\boldsymbol{y}, f(\boldsymbol{x})) \right] \tag{9}$$

Learning Capacity of a Function Class

Note that "population risk" does not depend on the choice of training data — even if the f being used is instantiated as the output of a training algorithm which consumed some particular training data. So, that motivates defining what the algorithm sees directly,

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Given a set of training data $\mathcal{S}_n = \{(\boldsymbol{x}_i, y_i), i = 1, \ldots, n\}$, each data being sampled from the distribution $\mathcal{D}$ specified above, we denote the empirical risk of f as,

$$\hat{R}(f, \mathcal{S}_n) \coloneqq \frac{1}{n} \sum_{i=1}^{n} \left[\ell(\boldsymbol{y}_i, f(\boldsymbol{x}_i))\right] \tag{10}$$

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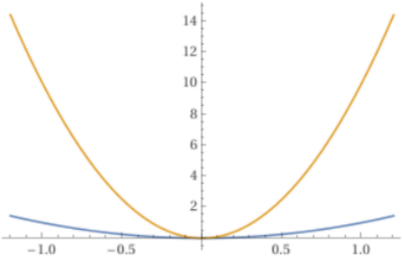
"Will taking increasingly large number of samples ensure that $\hat{R}(f, \mathcal{S}_n)$ converges to $R(f)$?" (Tricky!)

Learning Capacity of a Function Class

Given any loss function, *successful learning needs a very subtle interplay between* n *(the number of training data),* $\mathcal{D}$ *(the data distribution) and* $\mathcal{F}$ *(the model class)* which we shall begin to slowly uncover here.

— in particular we shall see how the "size of the predictors" i.e number of parameters in the specification of a predictor (like the number of weights in a neural net) *may not* directly affect this relationship!

Learning Capacity of a Function Class



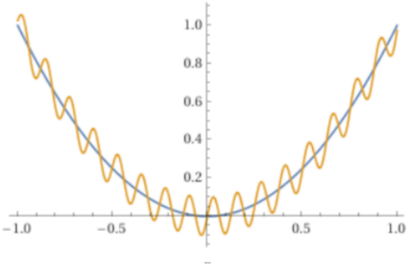


Figure 12: For a fixed choice of training samples, the yellow curves are possible cartoons for the values of the empirical risk for the predictors at the corresponding parameter values. The blue curves are possible depictions of the values of the population risk for the same parameter values. **This is the common lucky situation with many modern big models that the population and empirical risk are both nearly at their minima at the same set of parameters.**

Two Important Aspects of Common ML Algorithms

The Search Space of ML is Commonly Infinite.

(1) In trying to minimize $\hat{R}(f, \mathcal{S}_n)$, the set of f accessible to a algorithm is usually a parametric set of functions - like the set of neural nets at different weights on a fixed architecture. Thus the algorithm effectively searches through the set of f by searching through the space of parameters of it.

The space of all possible f i.e $\mathcal{F}$, that an algorithm is effectively searching through is very commonly an infinite dimensional space while the space of its parameters is not.

Two Important Aspects of Common ML Algorithms

Its a Very High Bar to be Successful ML Algorithm.

(2) After the search/optimization is over what we really want is that the obtained model — say $\mathrm{ALG}(\ell, \mathcal{S}_n, \mathcal{F})$, which is some function in the set $\mathcal{F}$ — have low predictive errors/losses on unseen data. But this unseen data is something that is randomly sampled from an (unknown to the algorithm) data distribution.

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Hence, we want to capture the idea that on-an-average the obtained model has low errors on *all possible data*.

This is captured by wanting that the population risk of the ML algorithm's output, $R(\mathrm{ALG}(\ell, \mathcal{S}_n, \mathcal{F}))$ be as low as possible— i.e we want $R(\mathrm{ALG}(\ell, \mathcal{S}_n, \mathcal{F})) \approx \inf_{f \in \mathcal{F}} R(f)$.

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Four Immediate Puzzles About ML

Puzzle A

Given a specific class of predictors $\mathcal{F}$, loss ℓ, data distribution $\mathcal{D}$ and a sample from it $\mathcal{S}_n$, **it is extremely difficult to figure out if the empirical and population risks are close for each predictor** (as in the earlier right side figure) — or if even the global minimas of the two risks, f_{erm} and $f_\star$ are closeby predictors/have closeby weights.

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Puzzle B

Its a further **hard challenge to figure out if the proximity between the two risks holds or not with high-probability over sampling the training data** —— recall that the shape of the empirical risk curve (the yellow line in the earlier figure) depends on the specific training data $\mathcal{S}_n$ chosen to define it.

Four Immediate Puzzles About ML

Puzzle C

Its natural to expect that with increasing number of training data being used, with high probability over sampling them, these two risk curves will eventually coincide in the large sample size limit.

But such a **large** n **convergence of** $\hat{R}(f, \mathcal{S}_n)$ **to** $R(f)$ **for all** $f \in \mathcal{F}$ **simultaneously, cannot be taken for granted** and establishing this is quite a technical quest to undertake. *(its a project suggestion!)* Such a convergence of the risks needs non-trivial properties to be true about the loss and the data and the predictors.

Four Immediate Puzzles About ML

Puzzle D

Recall the fact that the training data $\mathcal{S}_n$ is itself randomly sampled from the full distribution over data — and thus **any large n convergence of $\hat{R}(f, \mathcal{S}_n)$ to $R(f)$, for all $f \in \mathcal{F}$ simultaneously, also has to be with high probability over sampling $\mathcal{S}_n$.**

Always Remember :

Output of any ML algorithm is a random variable valued in the model space.

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Puzzle D

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Thus, we have no escape from having to live with the fact that "$\mathrm{ALG}(\ell, \mathcal{S}_n, \mathcal{F})$" — **the model output by the training algorithm — is itself random and its distribution is essentially unknowable. So there is an almost insurmountable challenge in being able to control the quantity that we are really after i.e $R(\mathrm{ALG}(\ell, \mathcal{S}_n, \mathcal{F}))$.**

Always Remember :

Output of any ML algorithm is a random variable valued in the model space.

What Can We Reasonably Hope to Measure in General?

The deeply fundamental idea that takes us beyond this conundrum will be to **relax our goals** from wanting to control the above **to want to control only the worst possible difference between population and empirical risk, in expectation over all possible training data $\mathcal{S}_n$.**

Stated precisely, the data averaged worst (over all possible predictors) "generalization gap" between the population and empirical risk is,

$$\mathbb{E}_{\mathcal{S}_n}\left[\sup_{f\in\mathcal{F}}\left(\underbrace{\hat{R}(f,\mathcal{S}_n) - R(f)}_{\textbf{Generalization Error/Gap of f}}\right)\right]$$

What Can We Reasonably Hope to Measure in General?

$$\mathbb{E}_{\mathcal{S}_n}\left[\sup_{f\in\mathcal{F}}\left(\hat{R}(f,\mathcal{S}_n) - R(f)\right)\right]$$

If we only try to control the above then that frees us from the details of any particular experiment that is trying to find "$\operatorname{arginf}_{f\in\mathcal{F}} R(f)$". In particular, by focusing thus we become blind to the details of any algorithm and data that might be deployed in trying to find $\operatorname{arginf}_{f\in\mathcal{F}} R(f)$.

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But this is the price we must pay to gain any insight at all into how the choice of $n, \mathcal{F}$ and $\mathcal{D}$ interact in our main goal towards finding,

$\operatorname{arginf}_{f\in\mathcal{F}} R(f)$

— *particularly when $\mathcal{F}$ is very complicated, like a set of deep neural nets!*

What Can We Reasonably Hope to Measure in General?

$$\mathbb{E}_{\mathcal{S}_n}\left[\sup_{f\in\mathcal{F}}\left(\hat{R}(f,\mathcal{S}_n) - R(f)\right)\right]$$

Thus **any condition we find on $n, \mathcal{F}$ and $\mathcal{D}$ which makes the above expectation small will be sufficient for the data averaged worst gap between population/"out-of-sample" risk for any predictor in the class $\mathcal{F}$ and its empirical risk, to be as much small.**

(it's curious that in the daily use-cases with large nets the empirical risk can often be made $\approx$ 0 !)

The Key Idea of Rademacher Complexity

Definition (Empirical Rademacher complexity)

For a function class $\mathcal{H}$, suppose being given a $n-$sized data-set $\mathcal{S}_n$ of points $\{\boldsymbol{z}_i \mid i = 1, \ldots, n\}$ in the domain of the elements in $\mathcal{H}$. Let $\boldsymbol{\epsilon}$ be a $n-$dimensional random vector s.t. its coordinates are sampled as $\epsilon_i \sim \pm 1$ with equal probability and independently. Then the corresponding empirical Rademacher complexity is defined as,

$$\hat{\mathcal{R}}_n(\mathcal{H}) \coloneqq \mathbb{E}_{\boldsymbol{\epsilon}}\left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \epsilon_i h(\boldsymbol{z}_i)\right].$$

The Key Idea of Rademacher Complexity

Definition (Average Rademacher complexity)

If the elements of $\mathcal{S}_n$ above are sampled independently and identically from a common distribution $\mathcal{D}$ then we shall denote this using the notation $\mathcal{S}_n \sim \mathcal{D}^n$. Then the "average Rademacher complexity" is defined as,

$$\mathcal{R}_n(\mathcal{H}) \coloneqq \mathbb{E}_{\mathcal{S}_n}\left(\mathbb{E}_{\boldsymbol{\epsilon}}\left[\sup_{h\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} \epsilon_i h(\boldsymbol{z}_i)\right]\right).$$

A Foundational Property of Rademacher Complexity

Now we are set to state a key property about Rademacher complexity,

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Theorem (**Rademacher Complexity & the Average of The Worst Generalization Gap)**

Suppose $\mathcal{H}$ is a set (also called a "class") of real valued functions on a common domain and let $\mathcal{S}_n$ denote n–sized samples being drawn from a distribution $\mathcal{D}$ on that domain. Then we have,

$$\mathbb{E}_{\mathcal{S}_n}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{z}_i)-\mathbb{E}_{\boldsymbol{z}}\left[h(\boldsymbol{z})\right]\right)\right]\leq 2\mathcal{R}_n(\mathcal{H})$$

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USECASE : If $\mathcal{H} = \ell \circ \mathcal{F}$ for some class of predictors $\mathcal{F}$, we see that above is a bound on the expectation (over training data) of the worst generalization error among all possible predictors – an upperbound that is dependent on the loss ℓ, the predictor class $\mathcal{F}$, the distribution $\mathcal{D}$, and the number of samples n.

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$$\mathbb{E}_{S_n}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n} h(z_i) - \mathbb{E}_z\left[h(z)\right]\right)\right] \leq 2\mathcal{R}_n(\mathcal{H})$$

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Being able to incorporate all the dependencies — on both sides of the equation — and for arbitrary model classes $\mathcal{F}$ — in such a succinct inequality as above makes this theorem such an attractive approach to understanding machine learning.

(most beautiful equation in ML? 😄)

How Much Training Data is Sufficient for Learning?

We shall often hope to find a function $\mathcal{C}(\mathcal{H}, \mathcal{D})$ s.t we can write,

$$\mathcal{R}_n(\mathcal{H}) \leq \frac{\mathcal{C}(\mathcal{H}, \mathcal{D})}{\sqrt{n}}$$

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To the best of our knowledge there is no simple general principle for when such an inequality as above is possible. Its kind of fortuitous that so many of the natural use-cases in machine learning practice have an inequality of the above form being true! Consequentially, we would have,

$$\mathbb{E}_{\mathcal{S}_n}\left[\sup_{h \in \mathcal{H}}\left(\hat{R}(h, \mathcal{S}_n) - R(h)\right)\right] \leq \frac{2 \cdot \mathcal{C}(\mathcal{H}, \mathcal{D})}{\sqrt{n}}$$

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We can read off from the above, that by training using $n \ge \left(\frac{2\cdot\mathcal{C}(\mathcal{H},\mathcal{D})}{\epsilon}\right)^2$ samples we can have the data averaged worst/maximum generalization error be at most ϵ, for any $\epsilon > 0$ arbitrarily small. **One of the immediate roles of the idea of Rademacher complexity is to be able to provide such "sample complexity" estimates i.e estimates of the #samples being sufficient to attain a target accuracy of learning.**

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As the function class $\mathcal{H}$ gets more and more complicated our sample complexity estimates from the above machinery often start to wildly overestimate the number of required training data than what we see to be needed in practice. Resolving this theory vs experiment gap is a key direction at the cutting-edge of research!

How Much Training Data is Sufficient for Learning?

$$\mathbb{E}_{\mathcal{S}_n}\left[\sup_{h\in\mathcal{H}}\Big(\hat{R}(h,\mathcal{S}_n) - R(h)\Big)\right] \leq \frac{2\cdot\mathcal{C}(\mathcal{H},\mathcal{D})}{\sqrt{n}}$$

If inequality of the above form is obtainable *and $\mathcal{C}$ not depend on the number of parameters in the functions in $\mathcal{F}$* then we can see how using a large number of parameters in our models may not hurt anything,

– seen via this lens.

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Then, in such examples, the effective "capacity" of the setup i.e $\mathcal{C}$ is controlled by properties of the $\mathcal{H}$ and $\mathcal{D}$ which can be set s.t. they do not scale with the number of parameters in $\mathcal{H}$ (and in turn $\mathcal{F}$) - i.e "Bigger Models Can Be Better" And this possibility motivates going forward with the program to understand the, "average of the worst generalization gap".

Proving The Key Property of Rademacher Complexity

Proof.

The proof uses a technique called "symmetrization" — which has far-reaching uses than just this. Given a data set $z_1, z_2, \ldots, z_n$, we imagine sampling another training data set $z'_1, z'_2, \ldots, z'_n$ s.t. as earlier, each is chosen independently of the other and from the same distribution $\mathcal{D}$.

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$$\begin{aligned}
&\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^{n} h(\boldsymbol{z}_i) - \mathbb{E}_{\boldsymbol{z}}\left[h(\boldsymbol{z}) \right] \right) \\
= &\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^{n} h(\boldsymbol{z}_i) - \mathbb{E}_{\boldsymbol{z}'_1, \boldsymbol{z}'_2, \ldots, \boldsymbol{z}'_n} \left[\frac{1}{n} \sum_{i=1}^{n} h(\boldsymbol{z}'_i) \right] \right)
\end{aligned} \tag{11}$$

□

Proving The Key Property of Rademacher Complexity

Proof.

$$\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{z}_i)-\mathbb{E}_{\boldsymbol{z}}\left[h(\boldsymbol{z})\right]\right)=\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{z}_i)-\mathbb{E}_{\boldsymbol{z}'_1,\boldsymbol{z}'_2,\ldots,\boldsymbol{z}'_n}\left[\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{z}'_i)\right]\right)$$

The key idea is that since the first term in the $\sup$ above and the second term therein depend on two different sets of data one can just take the expectation "$\mathbb{E}_{\boldsymbol{z}'_1,\boldsymbol{z}'_2,\ldots,\boldsymbol{z}'_n}$" also over the first term without changing its value.

$$\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{z}_i)-\mathbb{E}_{\boldsymbol{z}}\left[h(\boldsymbol{z})\right]\right)=\sup_{h\in\mathcal{H}}\left(\mathbb{E}_{\boldsymbol{z}'_1,\boldsymbol{z}'_2,\ldots,\boldsymbol{z}'_n}\left[\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{z}_i)-\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{z}'_i)\right]\right)$$

□

Proving The Key Property of Rademacher Complexity

Proof.

Now we notice a general property w.r.t any two variables x and y, the later being random, and a bivariate function g,

$$\sup_x (\mathbb{E}_y[g(x,y)]) \leq \sup_x \left(\mathbb{E}_y[\sup_{x'} g(x',y)] \right) = \mathbb{E}_y[\sup_x g(x,y)]$$

In the last equality above we realize that the outermost $\sup$ in the middle term has become redundant since nothing inside it anymore depends on x. Thus we can just remove it and rename the x' as x.

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Now we notice a general property w.r.t any two variables x and y, the later being random, and a bivariate function g,

$$\sup_x \left(\mathbb{E}_y[g(x,y)]\right) \leq \sup_x \left(\mathbb{E}_y[\sup_{x'} g(x',y)]\right) = \mathbb{E}_y[\sup_x g(x,y)]$$

In the last equality above we realize that the outermost $\sup$ in the middle term has become redundant since nothing inside it anymore depends on x. Thus we can just remove it and rename the x' as x. We use the above observation to get,

$$\sup_{h\in\mathcal{H}} \left(\frac{1}{n}\sum_{i=1}^{n} h(\boldsymbol{z}_i) - \mathbb{E}_{\boldsymbol{z}}\left[h(\boldsymbol{z})\right]\right) \leq \mathbb{E}_{\boldsymbol{z}_1',\boldsymbol{z}_2',\ldots,\boldsymbol{z}_n'}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n} h(\boldsymbol{z}_i) - \frac{1}{n}\sum_{i=1}^{n} h(\boldsymbol{z}_i')\right)\right]$$

□

Proving The Key Property of Rademacher Complexity

Proof.

Now we take expectation over the data $z_1, \ldots, z_n$ (on both sides of the previous equation) to get,

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{z}_1,\boldsymbol{z}_2,\ldots,\boldsymbol{z}_n}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n} h(\boldsymbol{z}_i) - \mathbb{E}_{\boldsymbol{z}}\left[h(\boldsymbol{z})\right]\right)\right] \\
\leq &\mathbb{E}_{\boldsymbol{z}_1,\boldsymbol{z}_2,\ldots,\boldsymbol{z}_n}\left[\mathbb{E}_{\boldsymbol{z}'_1,\boldsymbol{z}'_2,\ldots,\boldsymbol{z}'_n}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}(h(\boldsymbol{z}_i) - h(\boldsymbol{z}'_i))\right)\right]\right] \\
\leq &\mathbb{E}_{\substack{\boldsymbol{z}_1,\boldsymbol{z}_2,\ldots,\boldsymbol{z}_n \\ \boldsymbol{z}'_1,\boldsymbol{z}'_2,\ldots,\boldsymbol{z}'_n}}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}(h(\boldsymbol{z}_i) - h(\boldsymbol{z}'_i))\right)\right] \qquad (12)
\end{aligned}$$

□

Proving The Key Property of Rademacher Complexity

Proof.

Now we make another critical observation.

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Thus "$q(h)_i \coloneqq h(\boldsymbol{z}_i) - h(\boldsymbol{z}_i')$" (a new random variable) is a symmetric random variable i.e $q(h)_i$ and $-q(h)_i$ are equally likely - in an intuitive sense. Now suppose σ_i for $i = 1, \ldots, n$ are independent random variables s.t. each is ± 1 with equal probability.

Proving The Key Property of Rademacher Complexity

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Then we have,

$$\mathbb{E}_{\substack{z_1, z_2, \ldots, z_n \\ z_1', z_2', \ldots, z_n'}} \left[\mathbb{E}_{\sigma_1, \ldots, \sigma_n} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^{n} \sigma_i \cdot q(h)_i \right) \right] \right]$$

$$= \frac{1}{2^n} \sum_{\sigma_1 = \pm 1} \cdots \sum_{\sigma_n = \pm 1} \mathbb{E}_{\substack{z_1, z_2, \ldots, z_n \\ z_1', z_2', \ldots, z_n'}} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^{n} \sigma_i \cdot q(h)_i \right) \right]$$

Proving The Key Property of Rademacher Complexity

Proof.

- We have realized that the density at $q(h)_i$ and $-q(h)_i$ are the same — hence the value of each of the 2^n expectations in the RHS above is the same — irrespective of the values assumed by σ_is.

Proving The Key Property of Rademacher Complexity

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- Thus each term in the above summand is identical irrespective of the values assumed by the σ_is. Thus the introduction of these auxiliary σ random variables is redundant.

Proving The Key Property of Rademacher Complexity

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- We have realized that the density at $q(h)_i$ and $-q(h)_i$ are the same — hence the value of each of the 2^n expectations in the RHS above is the same — irrespective of the values assumed by σ_is.
- Thus each term in the above summand is identical irrespective of the values assumed by the σ_is. Thus the introduction of these auxiliary σ random variables is redundant. So, we can write,

□

Proving The Key Property of Rademacher Complexity

Proof.

$$\mathbb{E}_{\boldsymbol{z}_1,\boldsymbol{z}_2,\ldots,\boldsymbol{z}_n}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{z}_i)-\mathbb{E}_{\boldsymbol{z}}\left[h(\boldsymbol{z})\right]\right)\right]$$

seen earlier in equation 12 to be,

$$\leq\mathbb{E}_{\substack{\boldsymbol{z}_1,\boldsymbol{z}_2,\ldots,\boldsymbol{z}_n\\ \boldsymbol{z}'_1,\boldsymbol{z}'_2,\ldots,\boldsymbol{z}'_n}}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}(h(\boldsymbol{z}_i)-h(\boldsymbol{z}'_i))\right)\right]$$

by the argued redundancy of inserting the "fair coin tosses" σ

$$\leq\mathbb{E}_{\substack{\boldsymbol{z}_1,\boldsymbol{z}_2,\ldots,\boldsymbol{z}_n\\ \boldsymbol{z}'_1,\boldsymbol{z}'_2,\ldots,\boldsymbol{z}'_n}}\left[\mathbb{E}_{\sigma_1,\ldots,\sigma_n}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}\sigma_i\cdot(h(\boldsymbol{z}_i)-h(\boldsymbol{z}'_i))\right)\right]\right]$$

by sub-additivity of supremum,

$$\leq\mathbb{E}_{\substack{\boldsymbol{z}_1,\boldsymbol{z}_2,\ldots,\boldsymbol{z}_n\\ \boldsymbol{z}'_1,\boldsymbol{z}'_2,\ldots,\boldsymbol{z}'_n}}\left[\mathbb{E}_{\sigma_1,\ldots,\sigma_n}\left[\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}\sigma_i\cdot h(\boldsymbol{z}_i)\right)+\sup_{h\in\mathcal{H}}\left(\frac{1}{n}\sum_{i=1}^{n}-\sigma_i h(\boldsymbol{z}'_i)\right)\right]\right]$$

$\square$

Proving The Key Property of Rademacher Complexity

Proof.

by linearity,

$$\leq \mathbb{E}_{\substack{z_1, z_2, \ldots, z_n \\ z'_1, z'_2, \ldots, z'_n}} \left[\mathbb{E}_{\sigma_1, \ldots, \sigma_n} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^{n} \sigma_i \cdot h(z_i) \right) \right] \right]$$

$$+ \mathbb{E}_{\substack{z_1, z_2, \ldots, z_n \\ z'_1, z'_2, \ldots, z'_n}} \left[\mathbb{E}_{\sigma_1, \ldots, \sigma_n} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^{n} -\sigma_i \cdot h(z'_i) \right) \right] \right]$$

by evaluating the redundant expectations and renaming the σs,

$$\leq \mathbb{E}_{z_1, z_2, \ldots, z_n} \left[\mathbb{E}_{\epsilon} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^{n} \epsilon_i \cdot h(z_i) \right) \right] \right]$$

$$+ \mathbb{E}_{z'_1, z'_2, \ldots, z'_n} \left[\mathbb{E}_{\epsilon} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^{n} \epsilon_i \cdot h(z'_i) \right) \right] \right]$$

$$\leq 2 \cdot \mathcal{R}_n(\mathcal{H})$$

And thus the proof is done!